SPA617: RF Components Characterization – Part D: RF Amplifier Performance Analysis

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1 Aim of the Experiment

To experimentally analyze and characterize the linear and non-linear performance characteristics of an RF Low-Noise Amplifier (LNA) used in a radio astronomy receiver front-end:

1. Measuring small-signal power gain across a frequency band of 100–200 MHz.
2. Evaluating amplifier gain compression and identifying the exact input and output 1-dB compression points ($P_{\text{in, 1dB}}, P_{\text{out, 1dB}}$).
3. Performing a two-tone test with frequencies $f_1 = 10$ MHz and $f_2 = 13$ MHz to measure third-order intermodulation products ($2f_1 - f_2 = 7$ MHz and $2f_2 - f_1 = 16$ MHz) and extrapolate the Third-Order Intercept Point (TOI / IIP3 / OIP3).

2 Theory and Principles

2.1 Small-Signal Power Gain

An RF amplifier increases the amplitude of weak incoming signals from an antenna before down-conversion or digitizing. In its linear operating regime, the power gain is defined as:

$$\text{Gain (dB)} = P_{\text{out}} \text{ (dBm)} - P_{\text{in}} \text{ (dBm)} \tag{1}$$

where P_{in} is the incident power level supplied by the signal generator and P_{out} is the fundamental output power measured at a matched $50\,\Omega$ load.

2.2 Nonlinear Response, Gain Compression, and Harmonics

Active semiconductor devices exhibit non-linear transfer characteristics at higher drive powers. The output voltage response $y(t)$ can be modeled using a Taylor series expansion of the input signal $x(t)$:

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \dots \tag{2}$$

For a single-tone excitation $x(t) = A\cos(\omega t)$:

$$y(t) = \frac{a_2 A^2}{2} + \left(a_1 A + \frac{3a_3 A^3}{4}\right)\cos(\omega t) + \frac{a_2 A^2}{2}\cos(2\omega t) + \frac{a_3 A^3}{4}\cos(3\omega t) + \dots \tag{3}$$

Because $a_3 < 0$ for typical amplifying devices, the fundamental term $\left(a_1 A + \frac{3}{4}a_3 A^3\right)$ grows slower than linear expectations as A increases. The input power level at which the actual gain drops by 1.0 dB relative to the small-signal gain is the 1-dB Compression Point ($P_{\text{in, 1dB}}$):

$$\text{Gain}(P_{\text{in, 1dB}}) = \text{Gain}_{\text{small-signal}} - 1.0 \text{ dB} \tag{4}$$

2.3 Third-Order Intercept (TOI / IP3)

When two closely-spaced input tones f_1 and f_2 are applied simultaneously, higher-order cubic terms (a_3x^3) generate third-order intermodulation (IM3) products at frequencies:

$$f_{\text{IM3, lower}} = 2f_1 - f_2, \qquad f_{\text{IM3, upper}} = 2f_2 - f_1 \tag{5}$$

For $f_1 = 10$ MHz and $f_2 = 13$ MHz, the corresponding 3rd-order products appear at:

$$2f_1 - f_2 = 2(10) - 13 = 7 \text{ MHz} \tag{6}$$

$$2f_2 - f_1 = 2(13) - 10 = 16 \text{ MHz} \tag{7}$$

The fundamental power output grows with a $1:1$ slope (in dB scale), whereas the third-order non-linear terms scale with a $3:1$ slope.

Extrapolating these two linear responses yields a theoretical intersection point called the Third-Order Intercept (TOI / IP3). The corresponding input power is IIP3 and output power is OIP3.

3 Observation Tables and Quantitative Analysis

3.1 Part D1: Small-Signal Gain vs. Frequency

Reference Input Power (P_{in}): -20.0 dBm (constant)

Table 1: Amplifier small-signal power gain across frequency (100–200 MHz)

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	**Gain (dB)**
100	-20.0	2.78	**22.78**
110	-20.0	2.75	**22.75**
120	-20.0	2.68	**22.68**
130	-20.0	2.62	**22.62**
140	-20.0	2.53	**22.53**
150	-20.0	2.44	**22.44**
160	-20.0	2.40	**22.40**
170	-20.0	2.37	**22.37**
180	-20.0	2.34	**22.34**
190	-20.0	2.33	**22.33**
200	-20.0	2.32	**22.32**

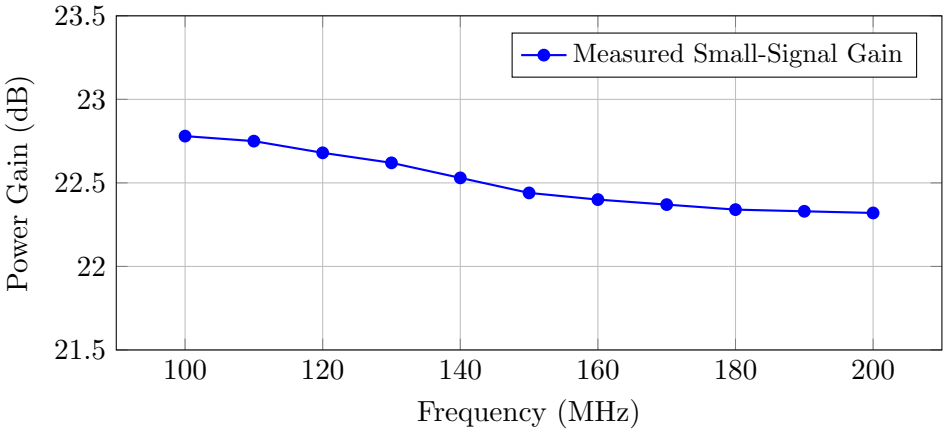


Figure 1: Amplifier power gain vs. frequency showing smooth passband response (≈ 22.5 dB baseline).

3.2 Part D2: Gain Compression and 1-dB Compression Point

Test Frequency (f_0): 100.0 MHz **Small-Signal Linear Gain ($\mathbf{Gain_{small\text{-}signal}}$):** 22.68 dB (at $P_{\text{in}} = -20$ dBm)
1-dB Compressed Gain Threshold: $22.68 - 1.00 = \mathbf{21.68}$ dB

Table 2: Power sweep at 100 MHz for gain compression measurement

P_{in} (dBm)	Measured P_{out} (dBm)	Actual Gain (dB)	**Gain Compression (dB)**
-20.0	2.68	22.68	0.00
-19.0	3.67	22.67	0.01
-18.0	4.64	22.64	0.04
-17.0	5.64	22.64	0.04
-16.0	6.62	22.62	0.06
-15.0	7.62	22.62	0.06
-14.0	8.60	22.60	0.08
-13.0	9.62	22.62	0.06
-12.0	10.58	22.58	0.10
-11.0	11.54	22.54	0.14
-10.0	12.52	22.52	0.16
-9.0	13.53	22.53	0.15
-8.0	14.42	22.42	0.26
-7.0	15.20	22.20	0.48
-6.0	15.90	21.90	0.78
-5.4	**16.28**	**21.68**	**1.00**
-5.0	16.47	21.47	1.21
-4.0	16.90	20.90	1.78
-3.0	17.20	20.20	2.48
-2.0	17.40	19.40	3.28
-1.0	17.54	18.54	4.14
0.0	17.61	17.61	5.07

1-dB Compression Evaluation: Fine tuning around $P_{\text{in}} \approx -5.4$ dBm gives an output power $P_{\text{out}} = +16.28$ dBm, corresponding to an actual gain of 21.68 dB (1.00 dB below the small-signal gain of 22.68 dB).

$$P_{\text{in, 1dB}} = \mathbf{-5.40} \text{ dBm}$$
$$P_{\text{out, 1dB}} = \mathbf{+16.28} \text{ dBm}$$

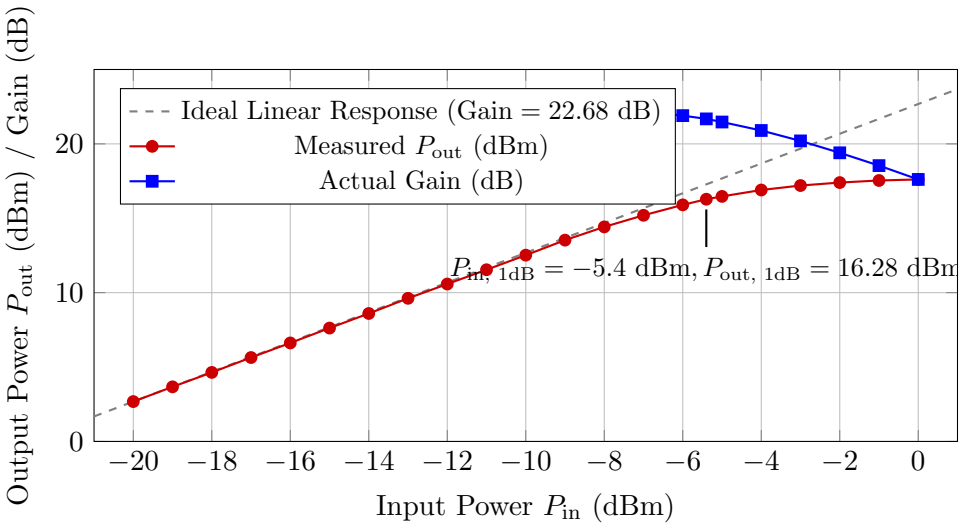


Figure 2: Amplifier Output Power and Gain vs. Input Power illustrating the 1-dB compression point at $P_{\text{in}} = -5.4$ dBm.

3.3 Part D3: Two-Tone Test and Third-Order Intercept (TOI)

Tone 1 Frequency (f_1): 10.0 MHz **Tone 2 Frequency (f_2):** 13.0 MHz
Lower IM3 Product ($2f_1 - f_2$): 7.0 MHz **Upper IM3 Product ($2f_2 - f_1$):** 16.0 MHz

Table 3: Fundamental and 3rd-order intermodulation output power levels

P_{in} (dBm)	Fundamental $P_{\text{out}}(f_1, f_2)$ (dBm)	3rd-Order Product $P_{\text{out}}(7, 16\text{ MHz})$ (dBm)	Fundamental Gain
-20.0	0.74	-61.20	20.74
-18.0	2.76	-57.60	20.76
-16.0	4.70	-51.20	20.70
-14.0	6.67	-45.09	20.67
-12.0	8.62	-38.25	20.62
-10.0	10.47	-25.07	20.47

Linear Fit and Intercept Derivation: Fitting the small-signal region ($P_{\text{in}} = -20$ to -14 dBm) gives:

$$P_{\text{out, fundamental}}(\text{dBm}) = 0.986 \cdot P_{\text{in}} + 20.49 \quad (\text{Slope} \approx 1:1)$$
$$P_{\text{out, 3rd-order}}(\text{dBm}) = 2.737 \cdot P_{\text{in}} - 7.25 \quad (\text{Slope} \approx 3:1)$$

Equating the two extrapolated linear trends:

$$0.986 \cdot \text{IIP3} + 20.49 = 2.737 \cdot \text{IIP3} - 7.25$$

$$\text{IIP3} = +\mathbf{15.85}\text{ dBm}$$

$$\text{OIP3} = \mathbf{0.986}(\mathbf{15.85}) + \mathbf{20.49} = +\mathbf{36.13}\text{ dBm}$$

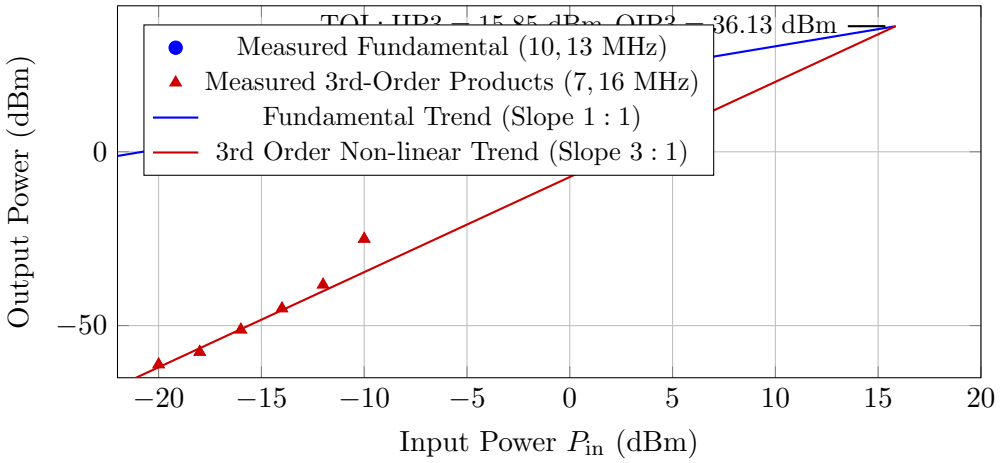


Figure 3: Third-Order Intercept (TOI) extrapolation showing the intersection of fundamental (1 : 1 slope) and third-order non-linear response (3 : 1 slope).

4 Summary of Experimental Results

Table 4: Principal Measured and Derived Amplifier Parameters

Parameter / Quantity	**Symbol**	**Measured / Calculated Value**
Small-Signal Power Gain (at 100 MHz)	$\text{Gain}_{\text{small-signal}}$	22.68 dB
Average Passband Gain (100–200 MHz)	Gain_{avg}	22.47 dB
Input 1-dB Compression Point	$P_{\text{in, 1dB}}$	-5.40 dBm
Output 1-dB Compression Point	$P_{\text{out, 1dB}}$	$+16.28$ dBm
Extrapolated Input 3rd-Order Intercept	IIP3	$+15.85$ dBm
Extrapolated Output 3rd-Order Intercept	OIP3	$+36.13$ dBm
OIP3 $-$ $P_{\text{out, 1dB}}$ Margin	$\Delta_{\text{TOI}-\text{1dB}}$	19.85 dB

5 Conclusions and Discussion

In this part of the experiment, an active RF amplifier was fully characterized across linear and non-linear regimes:

1. **Small-Signal Gain vs. Frequency:** Across 100–200 MHz, the small-signal gain remained stable between 22.32 dB and 22.78 dB (average gain $\approx$ 22.47 dB), showing excellent gain flatness (< 0.5 dB variation) across the band.

2. **1-dB Gain Compression:** Power sweep measurements at 100 MHz demonstrated linear amplification up to $P_{\text{in}} \approx -8.0$ dBm. The 1-dB compression point occurred at $P_{\text{in, 1dB}} = -5.40$ dBm and $P_{\text{out, 1dB}} = +16.28$ dBm.

3. **Two-Tone Test and Intercept Point:** Third-order intermodulation products resulting from input tones at $f_1 = 10$ MHz and $f_2 = 13$ MHz were measured at $2f_1 - f_2 = 7$ MHz and $2f_2 - f_1 = 16$ MHz. Fitting the linear $1:1$ fundamental slope and $3:1$ third-order non-linear slope yielded an extrapolated Output Third-Order Intercept of OIP3 $= +36.13$ dBm (IIP3 $= +15.85$ dBm).

4. **Physical Intercept Margin:** The measured OIP3 $-$ $P_{\text{out, 1dB}}$ difference was 19.85 dB, which closely aligns with the theoretical class-A amplifier expectation ($\approx$ 10–20 dB margin).